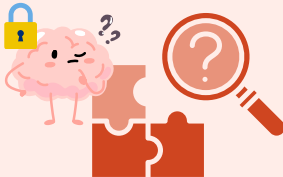


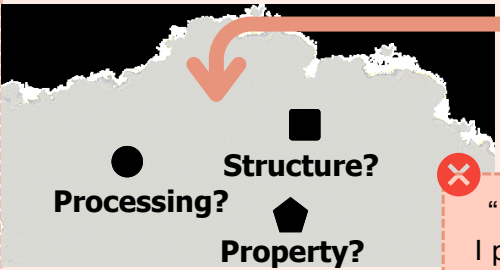


Protocol for synthesis of a high-T_c superconductor with La₂O₂Bi₃AgS₆

Naive KG+LLM



Rigid Retrieval



- Fragmented knowledge
- Fixed reasoning
- Unstable paths

Failure Output

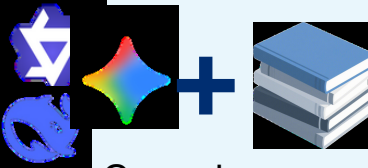
“KG info lacked ...
I propose using XX”



Hallucinated
synthesis,
not executable!



Protocol for synthesis of a high-T_c superconductor with La₂O₂Bi₃AgS₆



Causal-aware
Hierarchical Reasoning

Tier 1: Direct Match

Tier 2: Analogical

Tier 3: Fallback



- Causally integrated knowledge
- Adaptive hierarchical reasoning
- Robust paths

Successful Output

“725 °C solid-state;
high vacuum; 24–44 h
In₂S₃ dopant.”



SP-consistent,
experimentally
feasible.